NEET UG CHEMISTRY TUTORIAL

Organic Chemistry – Some Basic Principles and Techniques

Syllabus Context: Foundation of Organic Chemistry — Catenation, Hybridisation, Structural Representation, Classification, IUPAC Nomenclature, Isomerism, Electronic Displacements, Reaction Intermediates, Purification, and Qualitative & Quantitative Elemental Analysis.

1. Learning Objectives & Overview

After mastering this tutorial, you will be able to:

- Explain the tetravalency of carbon, orbital hybridisation (sp^3, sp^2, sp), molecular shapes, and the reactivity of π bonds.
- Convert seamlessly between Lewis, complete, condensed, bond-line, and 3-D wedge-and-dash structural formulas.
- Classify organic compounds based on carbon skeletons (acyclic, alicyclic, benzenoid, non-benzenoid, heterocyclic) and functional groups.
- Apply systematic IUPAC nomenclature rules to branched alkanes, polyfunctional compounds, and substituted benzene rings.
- Distinguish between structural isomerism (chain, position, functional, metamerism) and stereoisomerism.
- Analyze organic reaction mechanisms through homolytic and heterolytic bond fission, nucleophiles, electrophiles, and reaction intermediates (carbocations, carbanions, free radicals).
- Evaluate permanent and temporary electronic effects: Inductive (I), Resonance (R or M), Electromeric (E), and Hyperconjugation.
- Select appropriate purification methods (sublimation, crystallisation, simple/fractional/vacuum/steam distillation, differential extraction, and chromatography).
- Write chemical equations for qualitative elemental analysis (Lassaigne's test) and perform stoichiometric calculations for quantitative estimations (Liebig's, Dumas, Kjeldahl's, and Carius methods).

2. Tetravalence of Carbon & Shapes of Organic Molecules

2.1 Catenation and Historical Background

Carbon possesses the unique property of **catenation**—the ability to form stable covalent bonds with other carbon atoms, creating long chains, branched chains, and rings. It also forms strong covalent bonds with hydrogen, oxygen, nitrogen, sulphur, phosphorus, and halogens.

Around 1780, Swedish chemist Berzelius proposed the *'vital force theory'*, claiming that organic compounds could only be produced within living organisms. In 1828, **F. Wöhler** disproved this notion by synthesising an organic compound, **urea**, from an inorganic compound, **ammonium cyanate**, in the laboratory:

$$\underset{\text{Ammonium cyanate (Inorganic)}}{\mathrm{NH_4CNO}} \xrightarrow{\text{Heat}} \underset{\text{Urea (Organic)}}{\mathrm{NH_2CONH_2}}$$

Subsequent pioneering laboratory syntheses of **acetic acid by Kolbe (1845)** and **methane by Berthelot (1856)** conclusively proved that organic compounds can be synthesised from inorganic sources.

2.2 Hybridisation, Bond Length, Bond Enthalpy, and Electronegativity

The tetravalence of carbon and the geometries of organic molecules are governed by the hybridisation of its valence $2s$ and $2p$ orbitals:

- sp^3 **Hybridisation (e.g., Methane, $\mathrm{CH_4}$):** Four sp^3 hybrid orbitals (25% s-character, 75% p-character) form four σ bonds directed towards the corners of a regular **tetrahedron** (bond angle 109.5°).
- sp^2 **Hybridisation (e.g., Ethene, $\mathrm{C_2H_4}$):** Three sp^2 hybrid orbitals (33.3% s-character) lie in one plane forming a **trigonal planar** geometry (bond angle $\approx 120^\circ$), while one unhybridised p orbital lies perpendicular to the plane.
- sp **Hybridisation (e.g., Ethyne, $\mathrm{C_2H_2}$):** Two sp hybrid orbitals (50% s-character) orient at 180° in a **linear** geometry, with two mutually perpendicular unhybridised p orbitals.

Influence of Percentage s-Character

Because an s orbital is spherical and closer to the nucleus than a p orbital, increasing the s-character pulls the bonded electron pair closer to the carbon nucleus. Therefore, as s-character increases ($sp^3 < sp^2 < sp$):

- **Bond Length Decreases:** $\mathrm{C}(sp^3)\text{—}\mathrm{C}(sp^3)\ (154\,\mathrm{pm}) > \mathrm{C}(sp^2)\text{=}\mathrm{C}(sp^2)\ (134\,\mathrm{pm}) > \mathrm{C}(sp) \equiv \mathrm{C}(sp)$
- **Bond Enthalpy (Strength) Increases:** sp bonds $> sp^2$ bonds $> sp^3$ bonds
- **Electronegativity of Carbon Increases:** $\mathrm{C}(sp)\ (50\%\ s) > \mathrm{C}(sp^2)\ (33.3\%\ s) > \mathrm{C}(sp^3)\ (25\%\ s)$

2.3 Characteristic Features of π Bonds

In a π (pi) bond formation, parallel orientation of two unhybridised p orbitals on adjacent atoms is essential for proper sideways (lateral) overlap. In ethene ($\mathrm{H_2C = CH_2}$), all six atoms lie in the same plane, and the two unhybridised p orbitals are mutually parallel and perpendicular to the molecular plane.

- **Restricted Rotation:** Rotation of one $\mathrm{CH_2}$ fragment relative to the other disrupts the parallel overlap of the p orbitals. Hence, rotation about a carbon-carbon double bond ($\mathrm{C = C}$) is restricted.
- **Reactive Centres:** The electron charge cloud of a π bond is located above and below the plane of the bonding atoms. These loosely held π electrons are easily exposed and available to attacking electrophilic reagents, making multiple bonds the most reactive centres in unsaturated molecules.

3. Structural Representations of Organic Compounds

3.1 Complete, Condensed, and Bond-Line Formulas

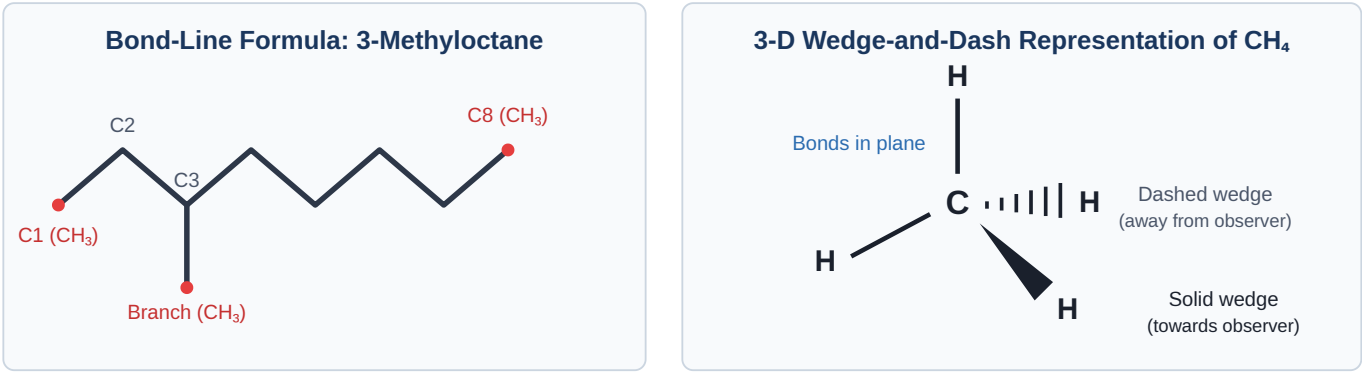
- **Complete Structural Formula (Dash Structure):** Every two-electron covalent bond is shown explicitly by a dash (single dash $-$ for single bond, double dash $=$ for double bond, triple dash $\equiv$ for triple bond). Lone pairs on heteroatoms (O, N, S, halogens) may or may not be shown.

- **Condensed Structural Formula:** Covalent dashes are omitted, and identical groups attached to an atom are indicated by a subscript. For example, octane is written as $CH_3(CH_2)_6CH_3$, ethane as CH_3CH_3, ethene as $H_2C = CH_2$, ethyne as $HC \equiv CH$, and methanol as CH_3OH.
- **Bond-Line Structural Formula:** Carbon and hydrogen atoms bonded to carbon are not written explicitly. Carbon-carbon bonds are drawn as zig-zag lines. Line terminals represent methyl ($-CH_3$) groups (unless a functional group is shown), and line junctions represent carbon atoms bonded to the appropriate number of hydrogens needed to satisfy carbon's tetravalency. Heteroatoms (e.g., O, N, Cl, Br) and hydrogens attached to heteroatoms (e.g., $-OH$, $-NH_2$) are written explicitly.

3.2 Three-Dimensional (Wedge-and-Dash) Representation and Molecular Models

To perceive the 3-D structure of an organic molecule on 2-D paper, wedge-and-dash conventions are used:

- **Normal line (—):** Bond lying in the plane of the paper.
- **Solid wedge (◀):** Bond projecting out of the plane of the paper, *towards* the observer (broad end towards observer).
- **Dashed wedge (·····):** Bond projecting out of the plane of the paper, *away* from the observer.

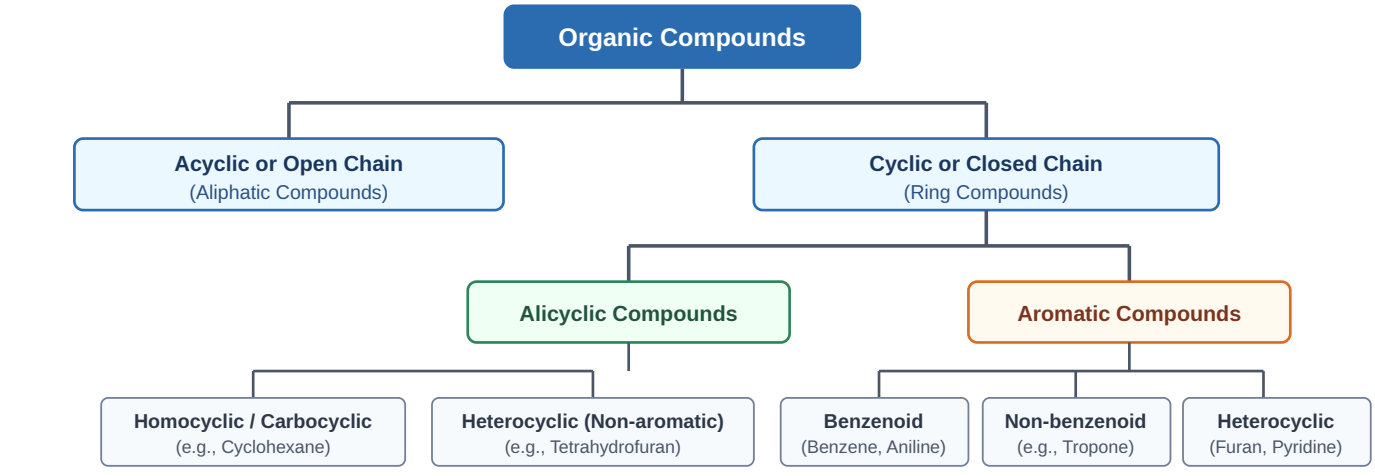


Tutorial Diagram 1: Left — Zig-zag bond-line representation of 3-methyloctane. Right — 3-D wedge-and-dash representation of methane.

Molecular Models: Physical devices used to visualise 3-D shapes include:

1. **Framework model:** Shows only the bonds connecting the atoms while ignoring atomic sizes.
2. **Ball-and-stick model:** Shows both atoms (balls) and bonds (sticks). For $C = C$ double bonds (like ethene), springs replace sticks (*ball-and-spring model*).
3. **Space-filling model:** Emphasises the relative size of each atom based on its van der Waals radius and conveys the volume occupied by each atom; bonds are not shown separately.

4. Classification of Organic Compounds



Tutorial Diagram 2: Structural classification of organic compounds.

- **Acyclic (Aliphatic) Compounds:** Straight or branched open-chain compounds such as ethane (CH_3CH_3), isobutane ($(CH_3)_2CHCH_3$), acetaldehyde (CH_3CHO), and acetic acid (CH_3COOH).
- **Alicyclic Compounds:** Aliphatic cyclic compounds. They may be *homocyclic* (only carbon atoms in the ring, e.g., cyclopropane, cyclohexane, cyclohexene) or *heterocyclic non-aromatic* (containing a heteroatom like O, N, or S in the ring, e.g., **tetrahydrofuran**, a 5-membered saturated ring containing four CH_2 units and one O atom).
- **Aromatic Compounds:**
  - *Benzenoid:* Contain at least one benzene ring (e.g., benzene, aniline, naphthalene).
  - *Non-benzenoid:* Aromatic rings without a benzene ring, such as **tropone** (a 7-membered ring with three conjugated $C = C$ double bonds and a $C = O$ group).
  - *Heterocyclic aromatic:* Aromatic rings containing heteroatoms, such as **furan** (5-membered ring with O), **thiophene** (5-membered ring with S), and **pyridine** (6-membered aromatic ring with N).

- **Functional Group & Homologous Series:** A functional group is an atom or group of atoms joined to the carbon chain responsible for characteristic chemical properties (e.g., $-\mathrm{OH}$, $-\mathrm{CHO}$, $-\mathrm{COOH}$). A series of compounds with the same functional group and general molecular formula where successive members differ by a $-\mathrm{CH_2}-$ unit (molecular mass $14\ \mathrm{u}$) is called a **homologous series**.

5. IUPAC Nomenclature of Organic Compounds

5.1 Common (Trivial) Names

Traditional names based on origin or properties remain widely used (e.g., formic acid from Latin *formica* meaning red ant; citric acid from citrus fruits; Buckminsterfullerene for the $\mathrm{C_{60}}$ cluster). Essential trivial names are summarised below:

Formula / Structure	Common / Trivial Name	Formula / Structure	Common / Trivial Name
$\mathrm{CH_4}$	Methane	$\mathrm{CHCl_3}$	Chloroform
$\mathrm{H_3CCH_2CH_2CH_3}$	n-Butane	$\mathrm{CH_3COOH}$	Acetic acid
$\mathrm{(H_3C)_2CHCH_3}$	Isobutane	$\mathrm{C_6H_6}$	Benzene
$\mathrm{(H_3C)_4C}$	Neopentane	$\mathrm{C_6H_5OCH_3}$	Anisole
$\mathrm{H_3CCH_2CH_2OH}$	n-Propyl alcohol	$\mathrm{C_6H_5NH_2}$	Aniline
HCHO	Formaldehyde	$\mathrm{C_6H_5COCH_3}$	Acetophenone
$\mathrm{(H_3C)_2CO}$	Acetone	$\mathrm{CH_3OCH_2CH_3}$	Ethyl methyl ether

5.2 Unbranched Alkanes and Branched Alkyl Groups

Straight-chain saturated hydrocarbons end with the suffix *'-ane'*: Methane ($\mathrm{CH_4}$), Ethane ($\mathrm{C_2H_6}$), Propane ($\mathrm{C_3H_8}$), Butane ($\mathrm{C_4H_{10}}$), Pentane ($\mathrm{C_5H_{12}}$), Hexane ($\mathrm{C_6H_{14}}$), Heptane ($\mathrm{C_7H_{16}}$), Octane ($\mathrm{C_8H_{18}}$), Nonane ($\mathrm{C_9H_{20}}$), Decane ($\mathrm{C_{10}H_{22}}$), Icosane ($\mathrm{C_{20}H_{42}}$), and Triacontane ($\mathrm{C_{30}H_{62}}$).

Removing one hydrogen atom from an alkane gives an **alkyl group** (methyl $-\mathrm{CH_3}$, ethyl $-\mathrm{CH_2CH_3}$, propyl $-\mathrm{CH_2CH_2CH_3}$, butyl $-\mathrm{CH_2CH_2CH_2CH_3}$, decyl $-\mathrm{CH_2(CH_2)_8CH_3}$). Important branched alkyl groups are:

- **Isopropyl:** $-\mathrm{CH(CH_3)_2}$
- ***sec*-Butyl:** $-\mathrm{CH(CH_3)CH_2CH_3}$
- **Isobutyl:** $-\mathrm{CH_2CH(CH_3)_2}$
- ***tert*-Butyl:** $-\mathrm{C(CH_3)_3}$
- **Neopentyl:** $-\mathrm{CH_2C(CH_3)_3}$

5.3 Rules for Naming Branched-Chain Alkanes and Cyclic Hydrocarbons

1. **Longest Chain Rule:** Identify the longest continuous carbon chain as the parent chain. If two chains of equal length compete, select the chain with the **greater number of side chains (substituents)**.
2. **Lowest Locant Rule:** Number the parent chain from the end that gives the substituted carbons the lowest possible locants (e.g., 2,5,6-trimethyloctane, not 3,4,7-trimethyloctane).
3. **Alphabetical Order of Prefixes:** List different alkyl substituents alphabetically before the parent alkane name. Separate numbers by commas and numbers from words by hyphens.
4. **Multiplying Prefixes & Trivial Alkyl Prefixes:**
   - Prefixes *di-, tri-, tetra-, penta-, hexa-* are **NOT** considered when alphabetising (e.g., *3-ethyl-4,4-dimethylheptane*: 'e' of ethyl precedes 'm' of methyl).
   - Prefixes ***iso-*** and ***neo-* ARE considered** part of the fundamental alkyl name for alphabetisation, whereas italicized ***sec-*** and ***tert-* are NOT considered** (e.g., in *5-sec-butyl-4-isopropyldecane*, 'b' of *sec*-butyl comes before 'i' of isopropyl).
5. **Equivalent Positions:** If two different substituents are at equivalent positions from either end, assign the **lower number to the one coming first alphabetically** (e.g., *3-ethyl-6-methyloctane*, not 6-ethyl-3-methyloctane).
6. **Complex (Branched) Substituents:** Number the side chain starting with **carbon-1 as the carbon directly attached to the parent root chain**, and enclose the complex branch name in parentheses, e.g., *5-(2-ethylbutyl)-3,3-dimethyldecane* or *5-(2,2-dimethylpropyl)nonane*.
7. **Cyclic Compounds:** Prefix *'cyclo'* to the alkane name. A more branched ring carbon gets the lower number (e.g., *3-ethyl-1,1-dimethylcyclohexane*, not 1-ethyl-3,3-dimethylcyclohexane). If substituents have equal locants, number alphabetically (e.g., *1-methyl-3-propylcyclohexane*).

5.4 Nomenclature of Compounds Containing Functional Groups

In polyfunctional compounds, select the **principal functional group** (which determines the main suffix and gets the lowest possible locant) according to the decreasing priority order:

$$-\mathrm{COOH} > -\mathrm{SO_3H} > -\mathrm{COOR} > -\mathrm{COCl} > -\mathrm{CONH_2} > -\mathrm{CN} > -\mathrm{HC{=}O} > > \mathrm{C{=}O} > -\mathrm{OH} > -\mathrm{NH_2} > > \mathrm{C{=}C} < > -\mathrm{C{\equiv}C}-$$

Groups such as $-\mathrm{R}$, $-\mathrm{C_6H_5}$ (phenyl), halogens ($-\mathrm{F}$, $-\mathrm{Cl}$, $-\mathrm{Br}$, $-\mathrm{I}$), $-\mathrm{NO_2}$ (nitro), and $-\mathrm{OR}$ (alkoxy) are **always treated as prefix substituents**. If multiple double or triple bonds occur, drop the terminal *'-ne'* of the alkane and add *'a'* (e.g., $\mathrm{CH_2{=}CH-CH{=}CH_2}$ is *buta-1,3-diene*). When an alkene and alkyne are both present, the chain is named as *alken-yne* (e.g., $\mathrm{CH{\equiv}C-CH{=}CH-CH{=}CH_2}$ is *hexa-1,3-dien-5-yne*).

Class of Compounds	Functional Group	IUPAC Prefix (Subordinate)	IUPAC Suffix (Principal)	Illustrative IUPAC Example
Carboxylic acids	$-\mathrm{COOH}$	carboxy	-oic acid	Butanoic acid, $\mathrm{CH_3(CH_2)_2CO_2H}$
Sulphonic acids	$-\mathrm{SO_3H}$	sulpho	sulphonic acid	Methylsulphonic acid, $\mathrm{CH_3SO_3H}$
Esters	$-\mathrm{COOR}$	alkoxycarbonyl	-oate	Methyl propanoate, $\mathrm{CH_3CH_2COOCH_3}$
Acyl halides	$-\mathrm{COX}$	halocarbonyl	-oyl halide	Butanoyl chloride, $\mathrm{CH_3(CH_2)_2COCl}$
Amides	$-\mathrm{CONH_2}$	-carbamoyl	-amide	Butanamide, $\mathrm{CH_3(CH_2)_2CONH_2}$
Nitriles	$-\mathrm{C\equiv N}$	cyano	nitrile	Pentanenitrile, $\mathrm{CH_3(CH_2)_3CN}$
Aldehydes	$-\mathrm{CHO}$	formyl, or oxo	-al	Butanal, $\mathrm{CH_3(CH_2)_2CHO}$
Ketones	$>\mathrm{C=O}$	oxo-	-one	Butan-2-one, $\mathrm{CH_3CH_2COCH_3}$
Alcohols	$-\mathrm{OH}$	hydroxy-	-ol	Butan-2-ol, $\mathrm{CH_3CH_2CH(OH)CH_3}$
Amines	$-\mathrm{NH_2}, >\mathrm{NH}, >\mathrm{N}-$	amino-	-amine	Butan-2-amine, $\mathrm{CH_3CH(NH_2)CH_2CH_3}$
Alkenes / Alkynes	$>\mathrm{C=C<}\ /\ -\mathrm{C\equiv C}-$	–	-ene / -yne	But-1-ene / But-1-yne
Halides / Nitro / Ethers	$-\mathrm{X}/-\mathrm{NO_2}/-\mathrm{OR}$	halo- / nitro / alkoxy-	(always prefix)	1-Bromobutane / Ethoxyethane

5.5 Substituted Benzene Compounds

- **Monosubstituted:** Prefix the substituent to *benzene* (e.g., methylbenzene [toluene], methoxybenzene [anisole], aminobenzene [aniline], nitrobenzene, bromobenzene).
- **Disubstituted:** Number positions to give the lowest locants: 1,2- (*ortho* or *o*-), 1,3- (*meta* or *m*-), and 1,4- (*para* or *p*-).
- **Tri- or Higher Substituted:** Prefixes *o-, m-, p-* cannot be used. Follow the lowest locant rule (e.g., *1-chloro-2,4-dinitrobenzene*). If an accepted trivial base name (like phenol, aniline, anisole, or toluene) is used, the group defining the base name is assigned **C-1**, and the ring is numbered in the direction that gives the next substituent the lowest locant (e.g., *2-chloro-4-methylanisole*, *4-ethyl-2-methylaniline*, *3,4-dimethylphenol*).

6. Isomerism

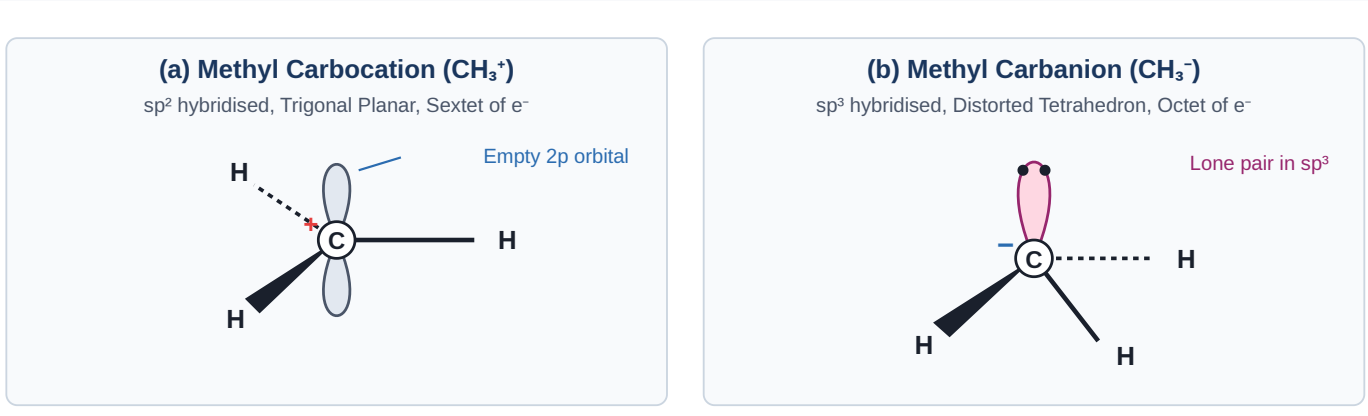
Compounds having the same molecular formula but different physical or chemical properties are called **isomers**.

1. **Structural Isomerism (Constitutional Isomerism):** Isomers differ in the connectivity (linking sequence) of atoms:
   - **Chain Isomerism:** Different carbon skeletons. Example: $\mathrm{C_5H_{12}}$ exists as *pentane* (straight chain), *isopentane* (2-methylbutane), and *neopentane* (2,2-dimethylpropane).
   - **Position Isomerism:** Same carbon skeleton and functional group, but different position of the substituent or functional group. Example: $\mathrm{C_3H_8O}$ gives *propan-1-ol* ($\mathrm{CH_3CH_2CH_2OH}$) and *propan-2-ol* ($\mathrm{CH_3CH(OH)CH_3}$).
   - **Functional Group Isomerism:** Different functional groups. Example: $\mathrm{C_3H_6O}$ represents *propanone* (ketone, $\mathrm{CH_3COCH_3}$) and *propanal* (aldehyde, $\mathrm{CH_3CH_2CHO}$).
   - **Metamerism:** Arises due to different alkyl chains on either side of a divalent/polyvalent functional group. Example: $\mathrm{C_4H_{10}O}$ represents *methoxypropane* ($\mathrm{CH_3OC_3H_7}$) and *ethoxyethane* ($\mathrm{C_2H_5OC_2H_5}$).
2. **Stereoisomerism:** Same constitution and covalent bond sequence, but different relative spatial arrangement of atoms or groups in 3-D space. Subdivided into **geometrical isomerism** (e.g., *cis/trans* across $\mathrm{C=C}$) and **optical isomerism**.

7. Fundamental Concepts in Organic Reaction Mechanism

An organic reaction involves an organic molecule (**substrate**, which supplies carbon to the new bond) reacting with an **attacking reagent** to form one or more reactive **intermediates** and finally **products** (plus byproducts). A step-by-step description of electron movement, energetics of bond cleavage/formation, and reaction kinetics is called the **reaction mechanism**.

7.1 Fission of a Covalent Bond and Reactive Intermediates



Tutorial Diagram 3: Shapes and orbital structures of (a) methyl carbocation and (b) methyl carbanion.

Feature	Carbocation (Carbonium Ion)	Carbanion	Free Radical
Mode of Cleavage	Heterolytic cleavage (shared pair leaves with the more electronegative atom)	Heterolytic cleavage (shared pair remains on carbon; Z^+ leaves)	Homolytic cleavage (one electron goes to each atom, shown by fish-hook arrow)
Charge & Electrons	Positively charged; **sextet** (6 valence e^-), diamagnetic	Negatively charged; **octet** (8 valence e^-, 1 lone pair), diamagnetic	Neutral; **7 valence** e^- (1 unpaired electron), paramagnetic
Hybridisation & Shape	sp^2 hybridised; **trigonal planar** with empty perpendicular $2p$ orbital	sp^3 hybridised; **distorted tetrahedron** (pyramidal)	Planar (sp^2) at radical centre
Order of Stability (Alkyl)	$\overset{+}{C}H_3 < CH_3\overset{+}{C}H_2 < (CH_3)_2\overset{+}{C}H < (CH_3)_3\overset{+}{C}$ (3° > 2° > 1° > methyl)	$\overset{-}{C}H_3 > CH_3\overset{-}{C}H_2 > (CH_3)_2\overset{-}{C}H > (CH_3)_3\overset{-}{C}$ (methyl > 1° > 2° > 3°;destabilised by $+I$ groups)	$\dot{C}H_3 < \dot{C}H_2CH_3 < \dot{C}H(CH_3)_2 < \dot{C}(CH_3)_3$ (3° > 2° > 1° > methyl)
Reaction Type	Ionic / Heteropolar / Polar	Ionic / Heteropolar / Polar	Free radical / Homopolar / Nonpolar (promoted by heat or light)

7.2 Nucleophiles and Electrophiles

- **Nucleophile ($Nu:$, "nucleus-seeking"):** An electron-rich reagent that brings an electron pair to an electron-deficient site (electrophilic centre) of the substrate.
  - *Negatively charged nucleophiles:* $HO^-, NC^-, HS^-, C_2H_5O^-$, carbanions ($R_3C:^-$).
  - *Neutral nucleophiles (with lone pairs):* $H_2\ddot{O}:, R_3N:, R_2NH, H_2N:^-$.
- **Electrophile (E^+, "electron-seeking"):** An electron-deficient reagent that takes away an electron pair from an electron-rich site (nucleophilic centre) of the substrate.
  - *Positively charged electrophiles:* Carbocations ($\overset{+}{C}H_3$), nitronium ion ($\overset{+}{N}O_2$), Cl^+, acylium ion ($CH_3\overset{+}{C}{=}O$).
  - *Neutral electrophiles:* Incomplete octet molecules like BF_3, or molecules with polar functional groups having an electrophilic carbon (δ^+) such as carbonyls ($>C=O$) and alkyl halides (R_3C-X).

7.3 Electron Displacement Effects in Covalent Bonds

(A) Inductive Effect (I Effect) — Permanent σ-Bond Polarisation

When a polar covalent bond (like $C-Cl$ in CH_3-CH_2-Cl) is present, carbon-1 acquires a partial positive charge (δ^+) and induces a smaller positive charge ($\delta\delta^+$) on carbon-2. This permanent polarisation of a σ bond caused by the polarisation of an adjacent σ bond is called the **inductive effect**. It diminishes rapidly with distance and becomes negligible after **three** carbon-carbon bonds.

- **Electron-Withdrawing Groups ($-I$ relative to H):** Halogens ($-F, -Cl, -Br, -I$), nitro ($-NO_2$), cyano ($-CN$), carboxy ($-COOH$), ester ($-COOR$), aryloxy ($-OAr$, e.g., $-OC_6H_5$).
- **Electron-Donating Groups ($+I$ relative to H):** Alkyl groups such as methyl ($-CH_3$), ethyl ($-CH_2CH_3$), isopropyl, and *tert*-butyl.

(B) Resonance and Resonance Effect (R or M Effect) — Permanent π-Delocalisation

When a single Lewis structure cannot explain the properties of a molecule, the actual molecule is a **resonance hybrid** of two or more hypothetical canonical (contributing) structures. For instance, experimental $C-C$ bond lengths in benzene are all uniform at $139\,pm$, intermediate between a $C-C$ single bond ($154\,pm$) and a $C=C$ double bond ($134\,pm$). The difference in energy between the actual resonance hybrid and the most stable canonical structure is the **resonance stabilisation energy**.

> **Rules for Evaluating Relative Stability of Resonance Structures**
>
> Canonical structures must have the same positions of nuclei and the same number of unpaired electrons. A canonical structure is **more stable** if it has:
>
> 1. More covalent bonds.
> 2. Complete octets on all atoms (duplet for H).
> 3. Less separation of opposite charges.
> 4. Any negative charge on a more electronegative atom and any positive charge on a more electropositive atom.
> 5. Greater dispersal of charge.

Resonance Effect (R or M): Polarity produced in a conjugated molecule by the interaction of two π bonds or between a π bond and a lone pair of electrons on an adjacent atom:

- $+R$ **(Positive Resonance) Effect:** Electron transfer is *away* from the substituent towards the conjugated system, increasing electron density at *ortho* and *para* positions (e.g., in aniline).
  $+R$ *Groups:* $-X$ (halogens), $-OH$, $-OR$, $-OCOR$, $-NH_2$, $-NHR$, $-NR_2$, $-NHCOR$.
- $-R$ **(Negative Resonance) Effect:** Electron transfer is *towards* the substituent away from the conjugated system, creating positive charges (δ^+) at *ortho* and *para* positions (e.g., in nitrobenzene).
  $-R$ *Groups:* $-COOH$, $-CHO$, $>C=O$, $-CN$, $-NO_2$.

(C) Electromeric Effect (E Effect) — Temporary Effect

Observed only in organic compounds having a **multiple bond** in the presence of an attacking reagent. It is the complete transfer of a shared pair of π electrons to one of the bonded atoms on the demand of an attacking reagent. When inductive and electromeric effects act in opposite directions, the **electromeric effect predominates**.

- $+E$ **Effect:** π electrons transfer to the atom to which the reagent attaches (e.g., addition of H^+ to $>C=C<$).
- $-E$ **Effect:** π electrons transfer to the atom to which the reagent does *not* attach (e.g., attack of CN^- on a multiple bond).

(D) Hyperconjugation — Permanent σ–π or σ–p Delocalisation ("No-Bond Resonance")

Hyperconjugation involves the delocalisation of σ electrons of a $C_\alpha-H$ bond of an alkyl group directly attached to an atom of an unsaturated system (alkenes, alkylarenes) or to an atom with an unshared empty/partially filled p orbital (carbocations, free radicals). The greater the number of α-$C-H$ bonds attached to the positively charged carbon or double bond, the greater the hyperconjugative stabilisation:

$$(CH_3)_3\overset{+}{C}\ (9\ \alpha\text{-H}) > (CH_3)_2\overset{+}{C}H\ (6\ \alpha\text{-H}) > CH_3\overset{+}{C}H_2\ (3\ \alpha\text{-H}) > \overset{+}{C}H_3\ (0\ \alpha\text{-H})$$

7.4 Types of Organic Reactions

Organic reactions are classified into four fundamental types: **(i) Substitution reactions** (an atom/group is replaced by another), **(ii) Addition reactions** (reagent adds across a multiple bond), **(iii) Elimination reactions** (loss of two atoms/groups to form a multiple bond), and **(iv) Rearrangement reactions** (reorganisation of the carbon skeleton or functional group position).

8. Methods of Purification of Organic Compounds

After extraction or synthesis, organic compounds are purified based on differences in physical properties. Purity is verified by sharp melting or boiling points, or via chromatographic/spectroscopic methods.

Purification Technique	Underlying Principle	Key Applications & Textbook Examples
1. Sublimation	Direct transition of a solid to vapour on heating without passing through the liquid state.	Separating sublimable solids (e.g., camphor, naphthalene, benzoic acid) from non-sublimable impurities (e.g., $CaSO_4$).
2. Crystallisation	Difference in solubilities of the compound and impurities in a suitable solvent (sparingly soluble at room temp, appreciably soluble at higher temp).	Purifying solid organic compounds. Coloured impurities are removed by adsorption over **activated charcoal**.
3. Simple Distillation	Liquids vaporise at different temperatures when they have a **sufficient difference in boiling points**, or to separate volatile liquids from non-volatile impurities.	Mixture of **chloroform (b.p. $334\ K$) and aniline (b.p. $457\ K$)**.
4. Fractional Distillation	Used when the difference in boiling points of two liquids is small. Repeated condensation and vaporisation occur in a **fractionating column** (each unit is a *theoretical plate*).	Separating different fractions of **crude oil in the petroleum industry**.
5. Distillation under Reduced Pressure (Vacuum Distillation)	A liquid boils when its vapour pressure equals external pressure. Lowering external pressure makes high-boiling liquids boil below their decomposition temperature.	Separating **glycerol from spent-lye in the soap industry**.
6. Steam Distillation	Used for substances that are **steam volatile and immiscible with water**. Mixture boils below $373\ K$ when $p = p_1\ (\text{organic}) + p_2\ (\text{water})$.	Separating **aniline from an aniline–water mixture**.
7. Differential Extraction	Shaking an aqueous solution with an immiscible organic solvent in which the organic compound is much more soluble.	Separating organic compounds from aqueous media using a separatory funnel (or continuous extraction if solubility is low).

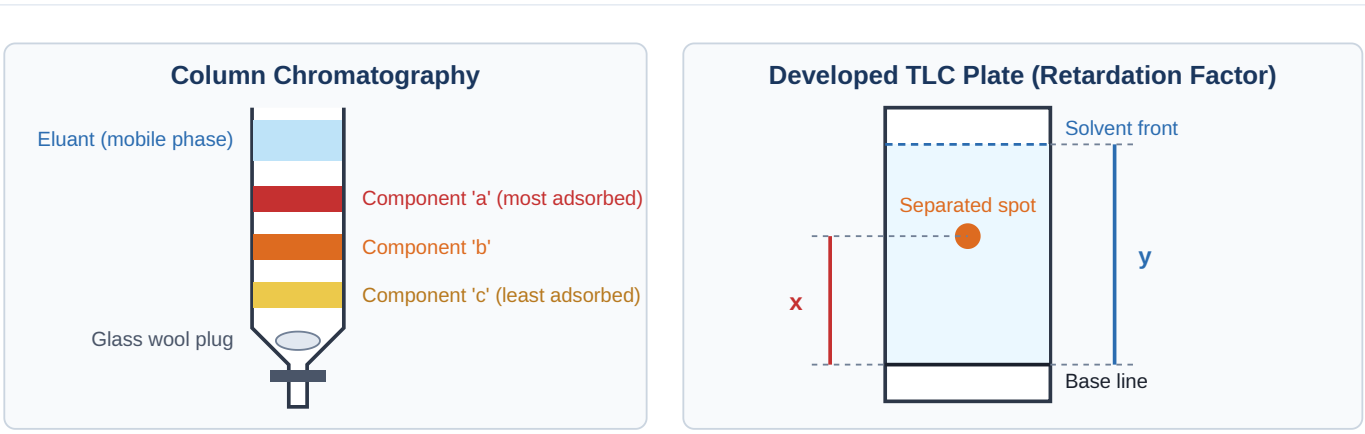
8.1 Chromatography

From the Greek *chroma* (colour). A mixture is applied to a **stationary phase** (solid or liquid) while a **mobile phase** (solvent, solvent mixture, or gas) moves over it:

1. **Adsorption Chromatography:** Based on differential adsorption of components onto a solid adsorbent (stationary phase: **silica gel** or **alumina**).
   - *Column Chromatography:* Adsorbent is packed in a vertical glass tube with a stopcock. The most strongly adsorbed component stays near the top; weakly adsorbed components elute first at the bottom.
   - *Thin Layer Chromatography (TLC):* Glass plate coated with a $0.2\ mm$ layer of silica gel/alumina (*chromaplate*). Sample is spotted $2\ cm$ above the base. As the eluant rises, components separate. Relative adsorption is measured by the **Retardation Factor** (R_f):

$$R_f = \frac{\text{Distance moved by the substance from base line } (x)}{\text{Distance moved by the solvent from base line } (y)}$$

   Detection of Colourless Spots on TLC: UV fluorescence, iodine crystals (brown spots), or spraying reagents (e.g., **ninhydrin solution** to detect amino acids).
2. **Partition Chromatography (Paper Chromatography):** Based on continuous differential partitioning of components between two liquid phases. Water trapped in chromatography paper acts as the **stationary phase**, while the ascending solvent is the **mobile phase**.



Tutorial Diagram 4: Differential adsorption in Column Chromatography (left) and measurement of x and y distances on a TLC chromatogram (right).

9. Qualitative & Quantitative Elemental Analysis

9.1 Qualitative Analysis (Detection of Elements)

- **Carbon and Hydrogen:** Heated with dry copper(II) oxide (CuO).
  - Carbon is oxidised to CO_2, which turns lime water milky: $CO_2 + Ca(OH)_2 \rightarrow CaCO_3 \downarrow + H_2O$.
  - Hydrogen is oxidised to H_2O, which turns white anhydrous $CuSO_4$ into blue hydrated $CuSO_4 \cdot 5H_2O$.

- **Lassaigne's Test (Sodium Fusion Extract for N, S, Halogens):** The organic compound is fused with metallic sodium to convert covalent elements into water-soluble ionic salts:

$$\mathrm{Na + C + N \xrightarrow{\Delta} NaCN} \quad ; \quad \mathrm{2Na + S \xrightarrow{\Delta} Na_2S} \quad ; \quad \mathrm{Na + X \xrightarrow{\Delta} NaX\ (X = Cl, Br, I)}$$

Element Tested	Reagents Added to Sodium Fusion Extract (SFE)	Key Chemical Equations & Diagnostic Observation
Nitrogen (N)	Boil with $\mathrm{FeSO_4}$, then acidify with conc. $\mathrm{H_2SO_4}$	$\mathrm{6CN^- + Fe^{2+} \rightarrow [Fe(CN)_6]^{4-}}$ $\mathrm{3[Fe(CN)_6]^{4-} + 4Fe^{3+} \xrightarrow{xH_2O} Fe_4[Fe(CN)_6]_3 \cdot xH_2O}$ **Observation: Prussian blue** colour (iron(III) hexacyanidoferrate(II) / ferriferrocyanide).
Sulphur (S) *(a) Lead acetate test* *(b) Nitroprusside test*	(a) Acidify with acetic acid + lead acetate (b) Sodium nitroprusside	(a) $\mathrm{S^{2-} + Pb^{2+} \rightarrow PbS\downarrow}$ (**Black precipitate**) (b) $\mathrm{S^{2-} + [Fe(CN)_5NO]^{2-} \rightarrow [Fe(CN)_5NOS]^{4-}}$ (**Violet colour**)
Both N and S together	$\mathrm{Fe^{3+}}$ salt	$\mathrm{Na + C + N + S \rightarrow NaSCN}$ $\mathrm{Fe^{3+} + SCN^- \rightarrow [Fe(SCN)]^{2+}}$ (**Blood red colour**; no Prussian blue due to absence of free $\mathrm{CN^-}$). *Note:* With excess Na, $\mathrm{NaSCN + 2Na \rightarrow NaCN + Na_2S}$.
Halogens (Cl, Br, I)	Acidify with conc. $\mathrm{HNO_3}$ (boil to expel HCN and $\mathrm{H_2S}$) + $\mathrm{AgNO_3}$	$\mathrm{X^- + Ag^+ \rightarrow AgX\downarrow}$ • AgCl: White ppt, **soluble** in $\mathrm{NH_4OH}$. • AgBr: Pale yellowish ppt, **sparingly soluble** in $\mathrm{NH_4OH}$. • AgI: Yellow ppt, **insoluble** in $\mathrm{NH_4OH}$.
Phosphorus (P)	Fuse with $\mathrm{Na_2O_2}$, boil with $\mathrm{HNO_3}$, add ammonium molybdate	$\mathrm{H_3PO_4 + 12(NH_4)_2MoO_4 + 21HNO_3 \rightarrow (NH_4)_3PO_4 \cdot 12MoO_3\downarrow + 21NH_4NO_3 + 12H_2O}$ **Observation: Canary yellow precipitate** of ammonium phosphomolybdate.

9.2 Quantitative Analysis (Estimation Formulas)

Let m (in grams) be the mass of the organic compound taken in each estimation:

1. Carbon & Hydrogen (Liebig's Combustion Method)

Compound burnt with excess $\mathrm{O_2}$ and CuO. Water (m_1 g) is absorbed in weighed anhydrous $\mathrm{CaCl_2}$, and $\mathrm{CO_2}$ (m_2 g) is absorbed in weighed conc. KOH solution:

$$\text{Percentage of Carbon} = \frac{12 \times m_2 \times 100}{44 \times m} \quad ; \quad \text{Percentage of Hydrogen} = \frac{2 \times m_1 \times 100}{18 \times m}$$

2. Nitrogen — (i) Dumas Method

Heated with CuO in $\mathrm{CO_2}$ atmosphere to yield free $\mathrm{N_2}$ gas (collected over aqueous KOH). Any nitrogen oxides formed are reduced back to $\mathrm{N_2}$ over heated copper gauze. If V_1 mL of $\mathrm{N_2}$ is collected at temperature T_1 K and atmospheric pressure P, then dry gas pressure $p_1 = P - \text{aqueous tension}$:

$$V\ (\text{at STP}) = \frac{p_1 V_1 \times 273}{760 \times T_1}\ \text{mL} \quad ; \quad \text{Percentage of Nitrogen} = \frac{28 \times V \times 100}{22400 \times m}$$

3. Nitrogen — (ii) Kjeldahl's Method

Heated with conc. $\mathrm{H_2SO_4}$ (in presence of $\mathrm{CuSO_4}$ catalyst and $\mathrm{K_2SO_4}$) to convert N into $\mathrm{(NH_4)_2SO_4}$, then distilled with excess NaOH to liberate $\mathrm{NH_3}$ into a known volume of standard acid:

$$\text{Percentage of Nitrogen} = \frac{1.4 \times M \times 2(V - V_1/2)}{m} = \frac{1.4 \times \text{Normality of acid neutralized} \times \text{Volume of acid (mL)}}{m}$$

Crucial Limitation: Kjeldahl's method is **NOT applicable** to nitro ($-\mathrm{NO_2}$), azo ($-\mathrm{N=N-}$), and ring-nitrogen compounds (e.g., **pyridine**, quinoline) because their nitrogen is not quantitatively converted to $\mathrm{(NH_4)_2SO_4}$.

4. Halogens, Sulphur, Phosphorus (Carius Method) & Oxygen

- **Halogens (as AgX, mass m_1 g):** $\%\,\mathrm{X} = \dfrac{\text{Atomic mass of X} \times m_1 \times 100}{\text{Molar mass of AgX} \times m}$ $(\mathrm{AgCl} = 143.5,\ \mathrm{AgBr} = 188,\ \mathrm{AgI} = 235\ \mathrm{g\,mol^{-1}})$
- **Sulphur (as $\mathrm{BaSO_4}$, mass m_1 g):** $\%\,\mathrm{S} = \dfrac{32 \times m_1 \times 100}{233 \times m}$
- **Phosphorus:** As ammonium phosphomolybdate $\mathrm{(NH_4)_3PO_4 \cdot 12MoO_3}$ ($M = 1877\ \mathrm{g\,mol^{-1}}$): $\%\,\mathrm{P} = \dfrac{31 \times m_1 \times 100}{1877 \times m}$.
  As magnesium pyrophosphate $\mathrm{Mg_2P_2O_7}$ ($M = 222\ \mathrm{g\,mol^{-1}}$): $\%\,\mathrm{P} = \dfrac{62 \times m_1 \times 100}{222 \times m}$.
- **Oxygen (Aluise's Method):** $\mathrm{O_2}$ converted to CO over red-hot coke at 1373 K, then oxidised by warm $\mathrm{I_2O_5}$ to $\mathrm{CO_2}$ (m_1 g): $\%\,\mathrm{O} = \dfrac{32 \times m_1 \times 100}{88 \times m}$.

10. Comprehensive Worked Examples

Worked Example 1: Sigma (σ) and Pi (π) Bond Counting & Hybridisation

Problem: Determine the number of σ and π bonds and the hybridisation of each carbon atom in: (a) $\mathrm{HC \equiv CCH = CHCH_3}$ (pent-3-en-1-yne) and (b) $\mathrm{CH_3CH = CHCN}$ (but-2-enenitrile).

Solution:

- **(a)** $\overset{1}{\mathrm{H}}\,\mathrm{C} \equiv \overset{2}{\mathrm{C}} - \overset{3}{\mathrm{C}}\,\mathrm{H} = \overset{4}{\mathrm{C}}\,\mathrm{H} - \overset{5}{\mathrm{C}}\,\mathrm{H_3}$:
  - $\mathrm{C-C}$ σ bonds $= 4$; $\mathrm{C-H}$ σ bonds $= 6$ (total 10 σ bonds).
  - π bonds: One $\pi_{\mathrm{C=C}}$ bond (between C3 and C4) + Two $\pi_{\mathrm{C\equiv C}}$ bonds (between C1 and C2) $= 3$ π bonds.
  - Hybridisation: C1 and C2 are sp; C3 and C4 are sp^2; C5 is sp^3.

- **(b)** $\overset{4}{\mathrm{C}}\mathrm{H_3}-\overset{3}{\mathrm{C}}\mathrm{H}=\overset{2}{\mathrm{C}}\mathrm{H}-\overset{1}{\mathrm{C}}\equiv\mathrm{N}$:
  - Hybridisation from left to right: $\mathrm{CH_3}$ carbon is sp^3; the two $\mathrm{CH}=\mathrm{CH}$ carbons are sp^2; the $-\mathrm{CN}$ carbon is sp.

Worked Example 2: Polyfunctional IUPAC Nomenclature

Problem: Write the systematic IUPAC names for the following compounds:
(a) $\mathrm{CH_3-CO-CH_2-CH_2-CH_2-COOH}$
(b) $\mathrm{CH_2=CH-CH_2-CH(OH)-CH_3}$
(c) A benzene ring having $-\mathrm{F}$ at position 1, $-\mathrm{NO_2}$ at position 2, and $-\mathrm{CH_2CH_3}$ at position 4.

Solution:

- **(a)** Contains a carboxylic acid ($-\mathrm{COOH}$) and a ketone ($>\mathrm{C=O}$). Since $-\mathrm{COOH}$ has higher priority than $>\mathrm{C=O}$, the suffix is *'-oic acid'* and the ketone is named as a prefix (*'oxo'*). Numbering from $-\mathrm{COOH}$ as C1 puts the keto group at C5 on a 6-carbon chain. **IUPAC Name: 5-Oxohexanoic acid**.
- **(b)** Contains an alcohol ($-\mathrm{OH}$) and an alkene ($>\mathrm{C=C}<$). Alcohol has higher priority than alkene, so number the 5-carbon chain from the right end to give $-\mathrm{OH}$ locant 2 and the double bond locant 4. **IUPAC Name: Pent-4-en-2-ol**.
- **(c)** All three groups ($-\mathrm{F}$, $-\mathrm{NO_2}$, $-\mathrm{C_2H_5}$) are prefix substituents. The locant set (1,2,4) is the lowest possible. Listing substituents alphabetically: **4-Ethyl-1-fluoro-2-nitrobenzene**.

Worked Example 3: Acidity and Alkoxide Stability via Electronic Effects

Problem: (a) Which ion is more stable: $\mathrm{O_2NCH_2CH_2O^-}$ or $\mathrm{CH_3CH_2O^-}$?
(b) Account for the acidity order: $\mathrm{Cl_3CCOOH} > \mathrm{Cl_2CHCOOH} > \mathrm{ClCH_2COOH}$.

Solution:

- **(a)** $\mathrm{O_2NCH_2CH_2O^-}$ is more stable than $\mathrm{CH_3CH_2O^-}$. The $-\mathrm{NO_2}$ group is a strong electron-withdrawing ($-I$) group that disperses the negative charge on the oxygen atom, thereby stabilising the alkoxide ion. Conversely, the $-\mathrm{CH_3}$ group exerts a $+I$ (electron-donating) effect that intensifies the negative charge and destabilises $\mathrm{CH_3CH_2O^-}$.
- **(b)** Chlorine exerts a $-I$ effect. As the number of chlorine atoms increases from one to three, the $-I$ inductive withdrawal increases, which disperses the negative charge of the conjugate carboxylate anion ($\mathrm{Cl_3CCOO^-}$) most effectively and facilitates $\mathrm{H^+}$ release.

Worked Example 4: Quantitative Estimation by Dumas and Kjeldahl's Methods

Problem:
(a) In Dumas' method, $0.30\,\mathrm{g}$ of an organic compound gave $50\,\mathrm{mL}$ of nitrogen collected at $300\,\mathrm{K}$ and $715\,\mathrm{mm\,Hg}$ pressure (aqueous tension at $300\,\mathrm{K} = 15\,\mathrm{mm\,Hg}$). Find $\%\,\mathrm{N}$.
(b) $0.50\,\mathrm{g}$ of an organic compound was analysed by Kjeldahl's method. The evolved $\mathrm{NH_3}$ was absorbed in $50\,\mathrm{mL}$ of $0.5\,\mathrm{M}$ $\mathrm{H_2SO_4}$. The residual acid required $60\,\mathrm{mL}$ of $0.5\,\mathrm{M}$ NaOH for neutralisation. Find $\%\,\mathrm{N}$.

Solution:

- **(a) Dumas Method:** Pressure of dry $\mathrm{N_2}$, $p_1 = 715 - 15 = 700\,\mathrm{mm\,Hg}$.

$$V_{\mathrm{STP}} = \frac{p_1 V_1 \times 273}{760 \times T_1} = \frac{700 \times 50 \times 273}{760 \times 300} = 41.9\,\mathrm{mL}$$

$$\%\,\mathrm{N} = \frac{28 \times V_{\mathrm{STP}} \times 100}{22400 \times m} = \frac{28 \times 41.9 \times 100}{22400 \times 0.30} = 17.46\%$$

- **(b) Kjeldahl's Method:**
  - Initial millimoles of $\mathrm{H^{Available}}$ from $\mathrm{H_2SO_4} = 2 \times M \times V = 2 \times 0.5 \times 50 = 50\,\mathrm{mmol}$.
  - Millimoles of $\mathrm{OH^-}$ used to neutralise residual acid $= 0.5 \times 60 = 30\,\mathrm{mmol}$.
  - Millimoles of $\mathrm{H^+}$ consumed by $\mathrm{NH_3}$ (= millimoles of $\mathrm{NH_3}$) $= 50 - 30 = 20\,\mathrm{mmol}$.
  - Mass of $\mathrm{N} = 20 \times 10^{-3}\,\mathrm{mol} \times 14\,\mathrm{g\,mol^{-1}} = 0.28\,\mathrm{g}$.
  - $\%\,\mathrm{N} = \dfrac{0.28}{0.50} \times 100 = 56.0\%$.

11. NEET UG Practice Questions with Worked Solutions

Note: The following are original practice questions designed strictly in accordance with the NEET UG examination standard to test conceptual clarity, nomenclature rules, electronic effects, purification, and quantitative analysis.

Practice Question 1 (Single-Choice MCQ)

In the organic compound $\overset{6}{\mathrm{C}}\mathrm{H}_2=\overset{5}{\mathrm{C}}\mathrm{H}-\overset{4}{\mathrm{C}}\mathrm{H}_2-\overset{3}{\mathrm{C}}\mathrm{H}_2-\overset{2}{\mathrm{C}}\equiv\overset{1}{\mathrm{C}}\mathrm{H}$ (hex-5-en-1-yne), the pair of hybridised orbitals involved in the formation of the $\mathrm{C}_2-\mathrm{C}_3$ bond is:

(1) $sp-sp^2$

(2) $sp-sp^3$

(3) sp^2-sp^3

(4) sp^3-sp^3

Correct Answer: (2)

Worked Solution: According to IUPAC numbering, a double bond takes precedence over a triple bond only when both receive the same locant from either end. Here, both are at terminal positions (locants 1 and 5), so numbering starts from the double bond: $\overset{1}{\mathrm{C}}\mathrm{H}_2=\overset{2}{\mathrm{C}}\mathrm{H}-\overset{3}{\mathrm{C}}\mathrm{H}_2-\overset{4}{\mathrm{C}}\mathrm{H}_2-\overset{5}{\mathrm{C}}\equiv\overset{6}{\mathrm{C}}\mathrm{H}$, or if numbered from the alkyne end as written in the formula ($\mathrm{C}_1\equiv\mathrm{C}_2-\mathrm{C}_3\mathrm{H}_2-$), let us check both conventions carefully: In standard textbook numbering from left to right ($\overset{1}{\mathrm{C}}\mathrm{H}_2=\overset{2}{\mathrm{C}}\mathrm{H}-\overset{3}{\mathrm{C}}\mathrm{H}_2-\overset{4}{\mathrm{C}}\mathrm{H}_2-\overset{5}{\mathrm{C}}\equiv\overset{6}{\mathrm{C}}\mathrm{H}$), C_2 is part of $=\mathrm{CH}-$ (sp^2) and C_3 is $-\mathrm{CH}_2-$ (sp^3), giving sp^2-sp^3. However, in the explicitly numbered formula given in the question ($\overset{6}{\mathrm{C}}\mathrm{H}_2=\overset{5}{\mathrm{C}}\mathrm{H}-\overset{4}{\mathrm{C}}\mathrm{H}_2-\overset{3}{\mathrm{C}}\mathrm{H}_2-\overset{2}{\mathrm{C}}\equiv\overset{1}{\mathrm{C}}\mathrm{H}$), C_2 is triply bonded (sp hybridised) and C_3 is a saturated $-\mathrm{CH}_2-$ carbon (sp^3 hybridised). Thus, for the explicitly numbered $\mathrm{C}_2-\mathrm{C}_3$ bond shown, the overlap is $sp-sp^3$. (If unnumbered, numbering from left to right gives C_2 as sp^2 and C_3 as sp^3, i.e., sp^2-sp^3).

Practice Question 2 (Single-Choice MCQ)

What is the total number of σ (sigma) and π (pi) bonds present in one molecule of N-methylmethanamide ($\mathrm{HCONHCH_3}$)?

(1) $8\,\sigma$ and $1\,\pi$

(2) $7\,\sigma$ and $2\,\pi$

(3) $8\,\sigma$ and $2\,\pi$

(4) $9\,\sigma$ and $1\,\pi$

Correct Answer: (1)

Worked Solution: Expanding the complete structure of $\mathrm{H}-\mathrm{C}(=\mathrm{O})-\mathrm{NH}-\mathrm{CH_3}$:
• $\mathrm{C}-\mathrm{H}$ σ bonds: 1 (formyl H) $+3$ (methyl H) $=4$.
• $\mathrm{N}-\mathrm{H}$ σ bond: 1.
• $\mathrm{C}-\mathrm{N}$ σ bonds: 2 (one $\mathrm{OC}-\mathrm{NH}$ bond and one $\mathrm{HN}-\mathrm{CH_3}$ bond).
• $\mathrm{C}=\mathrm{O}$ double bond: 1 σ bond and 1 π bond.
Total σ bonds $=4+1+2+1=8\,\sigma$ bonds, and $1\,\pi$ bond.

Practice Question 3 (Single-Choice MCQ)

What is the correct IUPAC name of the following polyfunctional organic compound?

$$\mathrm{HOCH_2-CH_2-CH_2-CH_2-CH_2-CO-CH_3}$$

(1) 2-Oxoheptan-7-ol

(2) 7-Hydroxyheptan-2-one

(3) 1-Hydroxyheptan-6-one

(4) 6-Oxoheptan-1-ol

Correct Answer: (2)

Worked Solution: In the functional group priority order, a ketone ($>\mathrm{C}=\mathrm{O}$) has higher priority than an alcohol ($-\mathrm{OH}$). Therefore, the ketone is the principal functional group (suffix *'-one'*) and the hydroxyl group is a subordinate substituent (prefix *'hydroxy-'*). Numbering the 7-carbon chain from the right end gives the carbonyl carbon locant 2 and the $-\mathrm{OH}$ group locant 7: **7-hydroxyheptan-2-one**.

Practice Question 4 (Single-Choice MCQ)

Consider a decane chain carrying an isopropyl group at C-4 and a *sec*-butyl group at C-5. What is its correct IUPAC name?

(1) 4-Isopropyl-5-*sec*-butyldecane

(2) 5-*sec*-Butyl-4-isopropyldecane

(3) 6-*sec*-Butyl-7-isopropyldecane

(4) 7-Isopropyl-6-*sec*-butyldecane

Correct Answer: (2)

Worked Solution: Numbering from the end closer to the first branch gives locants (4,5) rather than (6,7). When alphabetising trivial alkyl prefixes, *iso-* and *neo-* are considered part of the fundamental name, whereas italicized *sec-* and *tert-* are ignored. Thus, *sec*-**b**utyl is alphabetised under **'b'** and **i**sopropyl under **'i'**. Since 'b' comes before 'i', the name is **5-*sec*-butyl-4-isopropyldecane**.

Practice Question 5 (Single-Choice MCQ)

Which of the following pairs of compounds represents a pair of **metamers**?

(1) Propan-1-ol and Propan-2-ol

(2) Propanone and Propanal

(3) Methoxypropane ($CH_3OC_3H_7$) and Ethoxyethane ($C_2H_5OC_2H_5$)

(4) 2-Methylbutane and 2,2-Dimethylpropane

Correct Answer: (3)

Worked Solution: Metamerism arises when different alkyl chains are attached on either side of a divalent functional group (such as an ether linkage $-O-$ or ketone $-CO-$). Methoxypropane ($CH_3-O-CH_2CH_2CH_3$) and ethoxyethane ($CH_3CH_2-O-CH_2CH_3$) have the same molecular formula ($C_4H_{10}O$) and ether functional group, but different alkyl groups attached to oxygen.

Practice Question 6 (Statement-Based MCQ)

Examine the following two statements regarding nucleophiles and electrophiles:

Statement I: BF_3, $CH_3\overset{+}{C}=O$, and $\overset{+}{N}O_2$ act as electrophiles because their reactive sites are electron-deficient and can accept an electron pair.
Statement II: Neutral molecules such as $(CH_3)_3N:$ and $H_2O:$ cannot act as nucleophiles because they do not carry a net negative charge.
In the light of the above statements, choose the correct option:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

Correct Answer: (3)

Worked Solution: Statement I is correct: BF_3, $CH_3\overset{+}{C}=O$, and $\overset{+}{N}O_2$ have electron-deficient reactive centres (six valence electrons) and act as electrophiles. Statement II is incorrect: neutral molecules having unshared lone pairs of electrons, such as $H_2\ddot{O}:$, $(CH_3)_3N:$, and R_2NH, readily donate an electron pair and act as nucleophiles.

Practice Question 7 (Single-Choice MCQ)

Which of the following carbocations is the **most stable**?

(1) $(CH_3)_3C-\overset{+}{C}H_2$

(2) $(CH_3)_3\overset{+}{C}$

(3) $CH_3CH_2\overset{+}{C}H_2$

(4) $CH_3\overset{+}{C}HCH_2CH_3$

Correct Answer: (2)

Worked Solution: $(CH_3)_3\overset{+}{C}$ (*tert*-butyl cation) is a tertiary (3°) carbocation stabilised by **nine** α-$C-H$ **hyperconjugative structures** as well as the $+I$ inductive effect of three methyl groups. In contrast, $(CH_3)_3C-\overset{+}{C}H_2$ is a primary (1°) carbocation with *zero* α-$C-H$ bonds, $CH_3CH_2\overset{+}{C}H_2$ is primary with 2 α-H, and $CH_3\overset{+}{C}HCH_2CH_3$ is secondary with 5 α-H.

Practice Question 8 (Single-Choice MCQ)

Three resonance (canonical) structures of propenal ($CH_2=CH-CHO$) are given below:

$$CH_2=CH-\ddot{\underset{\cdot\cdot}{C}}H=\ddot{O}: \quad (I) \quad \longleftrightarrow \quad \overset{+}{C}H_2-CH=CH-\ddot{O}:^- \quad (II) \quad \longleftrightarrow \quad :\bar{C}H_2-CH=CH-\overset{+}{\ddot{O}}: \quad (III)$$

What is the correct order of relative stability of these contributing structures?

(1) $I > II > III$

(2) $II > I > III$

(3) $I > III > II$

(4) $III > II > I$

Correct Answer: (1)

Worked Solution: Structure **I** is the most stable because it has the maximum number of covalent bonds, complete octets on all carbon and oxygen atoms, and no charge separation. Between the charge-separated structures II and III, **II** is more stable than **III** because in II the negative charge resides on the more electronegative oxygen atom and the positive charge on the more electropositive carbon atom, whereas in III oxygen bears a positive charge with an incomplete octet.

Practice Question 9 (Single-Choice MCQ)

Which of the following sets contains **only** groups that exhibit a **negative resonance effect ($-R$ effect)** when attached to a conjugated system?

(1) $-COOH,\ -CHO,\ -CN,\ -NO_2$

(2) $-OH,\ -OR,\ -NH_2,\ -NO_2$

(3) $-Cl,\ -CHO,\ -CN,\ -NHCOR$

(4) $-COOH,\ -OCOR,\ -NR_2,\ >C=O$

Correct Answer: (1)

Worked Solution: Groups that withdraw π electron density from a conjugated system towards themselves exhibit the $-R$ effect. These typically have an atom attached to the conjugated system that is doubly or triply bonded to a more electronegative atom: $-COOH$, $-CHO$, $>C=O$, $-CN$, and $-NO_2$. Groups with lone pairs on the directly attached atom ($-OH$, $-OR$, $-NH_2$, $-NR_2$, $-NHCOR$, $-OCOR$, $-X$) exhibit the $+R$ effect.

Practice Question 10 (Matching MCQ)

Match the mixture/application in **Column I** with the most suitable purification technique in **Column II**:

Column I (Mixture / Application)	Column II (Purification Technique)
(A) Chloroform (b.p. $334\,\mathrm{K}$) and Aniline (b.p. $457\,\mathrm{K}$)	(i) Steam distillation
(B) Separation of crude oil fractions in petroleum industry	(ii) Distillation under reduced pressure
(C) Separation of glycerol from spent-lye in soap industry	(iii) Simple distillation
(D) Separation of aniline from aniline–water mixture	(iv) Fractional distillation

(1) A-(iii), B-(iv), C-(ii), D-(i)

(2) A-(iv), B-(iii), C-(i), D-(ii)

(3) A-(iii), B-(ii), C-(iv), D-(i)

(4) A-(i), B-(iv), C-(ii), D-(iii)

Correct Answer: (1)

Worked Solution:
• (A) Chloroform $(334\,\mathrm{K})$ and aniline $(457\,\mathrm{K})$ have a large boiling point difference $(123\,\mathrm{K})$ and are separated by **simple distillation (iii)**.
• (B) Crude oil fractions have close boiling points and are separated by **fractional distillation (iv)**.
• (C) Glycerol has a high boiling point and decomposes before boiling; it is separated from spent-lye by **distillation under reduced pressure (ii)**.
• (D) Aniline is steam-volatile and immiscible with water; it is separated from an aniline–water mixture by **steam distillation (i)**.

Practice Question 11 (Assertion–Reason MCQ)

Assertion (A): In Lassaigne's test for halogens, the sodium fusion extract is boiled with concentrated nitric acid before adding silver nitrate solution.
Reason (R): Concentrated nitric acid decomposes NaCN and $\mathrm{Na_2S}$ (if nitrogen and sulphur are present) into volatile HCN and $\mathrm{H_2S}$ gases; otherwise, $\mathrm{CN^-}$ and $\mathrm{S^{2-}}$ ions would form precipitates (AgCN and $\mathrm{Ag_2S}$) with $\mathrm{AgNO_3}$ and interfere with the halogen test.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution: If nitrogen or sulphur is also present in the organic compound, the sodium fusion extract contains NaCN and $\mathrm{Na_2S}$. On adding $\mathrm{AgNO_3}$ directly, $\mathrm{CN^-}$ forms a white precipitate of AgCN and $\mathrm{S^{2-}}$ forms a black precipitate of $\mathrm{Ag_2S}$, masking the silver halide (AgX) precipitate. Boiling with conc. $\mathrm{HNO_3}$ decomposes them prior to adding $\mathrm{AgNO_3}$.

Practice Question 12 (Single-Choice MCQ)

In Lassaigne's test, when both nitrogen and sulphur are present in an organic compound and sodium is not in excess, treating the sodium fusion extract with $\mathrm{Fe^{3+}}$ ions produces a blood-red colour due to the formation of:

(1) $\mathrm{Fe_4[Fe(CN)_6]_3 \cdot xH_2O}$

(2) $\mathrm{[Fe(CN)_5NOS]^{4-}}$

(3) $\mathrm{[Fe(SCN)]^{2+}}$

(4) $\mathrm{Na_4[Fe(CN)_6]}$

Correct Answer: (3)

Worked Solution: When both N and S are present, fusion with sodium forms sodium thiocyanate ($\mathrm{Na + C + N + S \rightarrow NaSCN}$). Since there are no free $\mathrm{CN^-}$ ions, Prussian blue ($\mathrm{Fe_4[Fe(CN)_6]_3}$) is not formed. Instead, $\mathrm{SCN^-}$ reacts with $\mathrm{Fe^{3+}}$ to give the **blood-red complex ion** $\mathrm{[Fe(SCN)]^{2+}}$.

Practice Question 13 (Single-Choice MCQ)

Kjeldahl's method cannot be used for the quantitative estimation of nitrogen in which of the following compounds?

(1) Urea (NH_2CONH_2)

(2) Aniline ($C_6H_5NH_2$)

(3) Pyridine (C_5H_5N)

(4) Ethanamine ($CH_3CH_2NH_2$)

Correct Answer: (3)

Worked Solution: Kjeldahl's method is not applicable to compounds containing nitrogen in **nitro** ($-NO_2$) groups, **azo** ($-N=N-$) groups, or nitrogen present in a **ring (such as pyridine)**, because the nitrogen in these compounds does not convert to ammonium sulphate ($(NH_4)_2SO_4$) on heating with concentrated H_2SO_4.

Practice Question 14 (Numerical MCQ)

An organic compound contains 69% carbon and 4.8% hydrogen, the remainder being oxygen. What are the masses of carbon dioxide (CO_2) and water (H_2O) produced when $0.20\ g$ of this substance is subjected to complete combustion?

(1) $0.506\ g\ CO_2$ and $0.0864\ g\ H_2O$

(2) $0.253\ g\ CO_2$ and $0.0432\ g\ H_2O$

(3) $0.506\ g\ CO_2$ and $0.1728\ g\ H_2O$

(4) $0.376\ g\ CO_2$ and $0.0864\ g\ H_2O$

Correct Answer: (1)

Worked Solution:

• From the carbon percentage formula: $\%\,C = \dfrac{12 \times m_{CO_2} \times 100}{44 \times m}$

$$m_{CO_2} = \frac{\%\,C \times 44 \times m}{12 \times 100} = \frac{69 \times 44 \times 0.20}{1200} = 0.506\,g$$

• From the hydrogen percentage formula: $\%\,H = \dfrac{2 \times m_{H_2O} \times 100}{18 \times m}$

$$m_{H_2O} = \frac{\%\,H \times 18 \times m}{2 \times 100} = \frac{4.8 \times 18 \times 0.20}{200} = 0.0864\,g$$

Practice Question 15 (Numerical MCQ)

In Carius estimation of halogens, $0.3780\ g$ of an organic chloro compound gave $0.5740\ g$ of silver chloride ($AgCl$). What is the percentage of chlorine present in the compound? (Given: Molar mass of $AgCl = 143.5\ g\,mol^{-1}$, Atomic mass of $Cl = 35.5\ u$)

(1) 18.78%

(2) 25.40%

(3) 37.57%

(4) 42.10%

Correct Answer: (3)

Worked Solution: Using the Carius formula for chlorine:

$$\text{Percentage of Chlorine} = \frac{35.5 \times m_{AgCl} \times 100}{143.5 \times m} = \frac{35.5 \times 0.5740 \times 100}{143.5 \times 0.3780}$$

$$\text{Notice that } \frac{0.5740}{143.5} = 0.0040\,mol \implies \%Cl = \frac{35.5 \times 0.0040 \times 100}{0.3780} = \frac{14.2}{0.3780} = 37.57\%$$

12. Common Pitfalls & NEET Revision Checklist

Frequent Exam Traps to Avoid

- **Trap 1 — Alphabetising Alkyl Prefixes:** Never ignore *iso-* and *neo-* during alphabetisation (they ARE part of the fundamental name), and never include *sec-*, *tert-*, *di-*, or *tri-* in alphabetisation.

- **Trap 2 — Alkene vs. Halide/Nitro Priority:** Halogens ($-\mathrm{X}$) and nitro ($-\mathrm{NO_2}$) groups are **always prefixes**; double and triple bonds take suffix priority over them. Thus, $\mathrm{BrCH_2CH=CH_2}$ is **3-bromoprop-1-ene** (NOT 1-bromoprop-2-ene), and a cyclohexene ring with $-\mathrm{NO_2}$ adjacent to the double bond is **3-nitrocyclohexene**.
- **Trap 3 — Acidification in Lead Acetate Test for Sulphur:** Always acidify the sodium fusion extract with **acetic acid** (and NOT $\mathrm{H_2SO_4}$) before adding lead acetate, because $\mathrm{H_2SO_4}$ would precipitate white $\mathrm{PbSO_4}$ even in the absence of sulphur!
- **Trap 4 — Direct $\mathrm{AgNO_3}$ Test on $\mathrm{CCl_4}$:** Heating $\mathrm{CCl_4}$ directly with $\mathrm{AgNO_3}$ does **not** give a white precipitate of AgCl because $\mathrm{CCl_4}$ is a covalent compound and does not ionise to release $\mathrm{Cl^-}$ ions (sodium fusion is required first).
- **Trap 5 — Subtracting Aqueous Tension in Dumas Method:** Always subtract the aqueous tension of water vapour from the recorded atmospheric pressure ($p_1 = P_{\text{atm}} - p_{\text{water}}$) before converting the collected $\mathrm{N_2}$ volume to STP.

Concise Final Recap

- **Electronegativity & Bond Strength:** sp $(50\%\ s) > sp^2$ $(33.3\%\ s) > sp^3$ $(25\%\ s)$.
- **Benzene $\mathrm{C-C}$ Bond Length:** $139\,\mathrm{pm}$ (between $\mathrm{C-C}$ $154\,\mathrm{pm}$ and $\mathrm{C=C}$ $134\,\mathrm{pm}$).
- **Principal Functional Group Priority:** $-\mathrm{COOH} > -\mathrm{SO_3H} > -\mathrm{COOR} > -\mathrm{COCl} > -\mathrm{CONH_2} > -\mathrm{CN} > -\mathrm{CHO} >> \mathrm{C=O} > -\mathrm{OH} > -\mathrm{NH_2} >> \mathrm{C=C} << -\mathrm{C\equiv C-}$.
- **Carbocation & Free Radical Stability:** $3^\circ > 2^\circ > 1^\circ > \mathrm{methyl}$ (governed by hyperconjugation and $+I$ effect).
- **Permanent vs. Temporary Effects:** Inductive (I), Resonance (R), and Hyperconjugation are permanent ground-state effects; Electromeric (E) is a temporary effect requiring an attacking reagent.
- **Key Lassaigne Colours:** Prussian blue $= \mathrm{Fe_4[Fe(CN)_6]_3 \cdot xH_2O}$ (N); Violet $= \mathrm{[Fe(CN)_5NOS]^{4-}}$ (S); Blood red $= \mathrm{[Fe(SCN)]^{2+}}$ (N + S); Canary yellow $= \mathrm{(NH_4)_3PO_4 \cdot 12MoO_3}$ (P).